

单液桥模型弯液面演化的数值模拟

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① 研究背景

② 研究方法

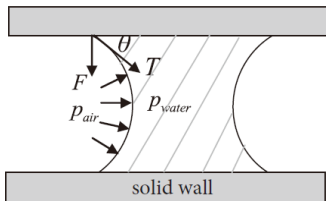
③ 结果讨论

④ 应用拓展

研究对象

单液桥模型

两个平行固壁之间的弯液面



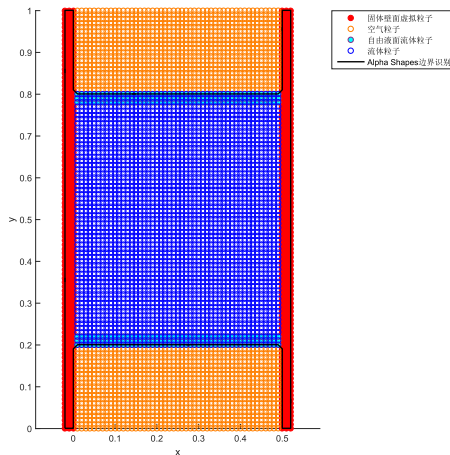
液桥力主要由表面张力与拉普拉斯压差引起！

θ : 固-液静态接触角

F : 附壁力

T : 表面张力

p : 压强



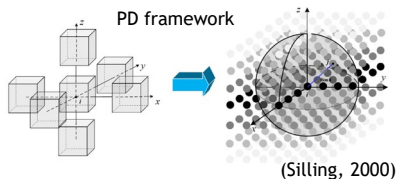
MATLAB 数值算例，初始的计算域被离散为2989个流体粒子、606个固体壁面虚构粒子和1960个空气粒子

研究现状

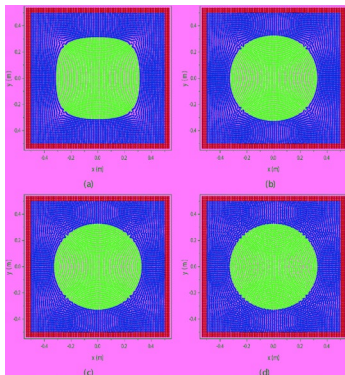
有关弯液面的数值建模方法

研究方法	模型	特点	局限性
DDA	一般模型	土体被离散为不规则的多边形颗粒	弯液面的固液接触角和曲率半径
	二维扩展模型	根据土体单位体积的含水量确定弯液面形态	恒定，不能获取动态演化过程
DEM	一般模型	表面张力的作用由黏附力代替	缺乏对弯液面的描述
	孤立毛细管模型	适用于低饱和度的二维土体	引入了众多参数来描述弯液面的形态，可能带来较大的计算误差，且不能获取动态演化过程
	索道液桥模型	适用于高饱和度的二维土体	
	复杂流体团簇模型	适用于高饱和度的三维土体	
CFD-DEM	单向流固耦合模型	采用CFD方法模拟水压，采用DEM方法模拟土体颗粒运动，考虑非饱和土中的基质吸力时采用恒定粘附力	流场不受土体颗粒运动的影响，不能完整地反映表面张力作用下的流固相互作用
MD	一般模型	纳米尺度的液桥模型	基于分子力场进行建模和计算，缺乏对表面张力的描述

近场动力学在流固耦合研究中的应用

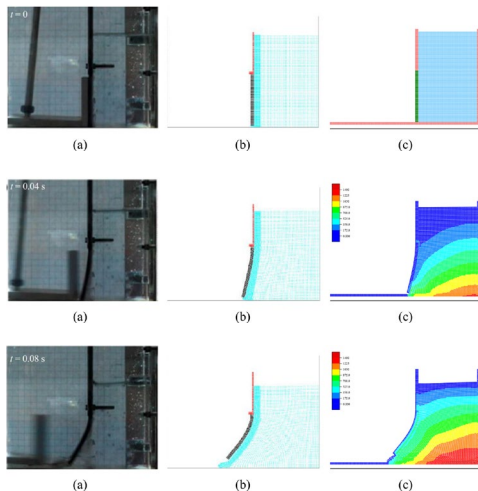


2D square droplet deformation simulation



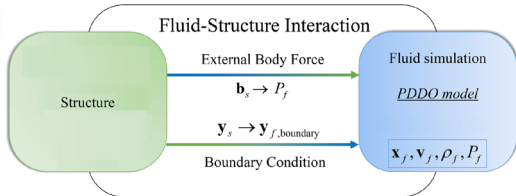
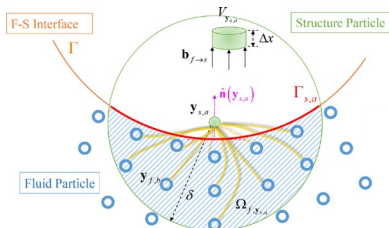
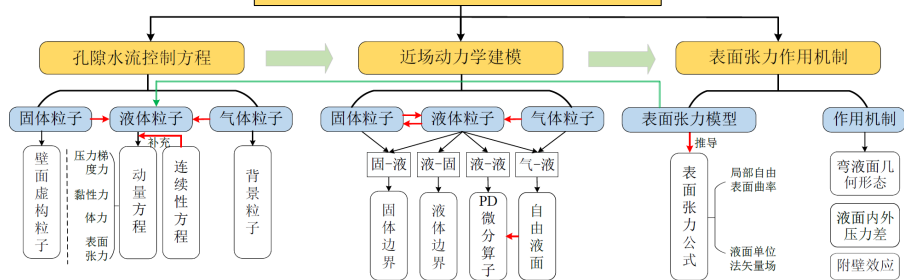
(Gao & Oterkus, 2020a)

Dam collapse under an elastic rubber gate



(Gao & Oterkus, 2020b)

单液桥模型弯液面演化的数值模拟



难点：计算复杂度高、规模大，边界条件施加困难！

① 研究背景

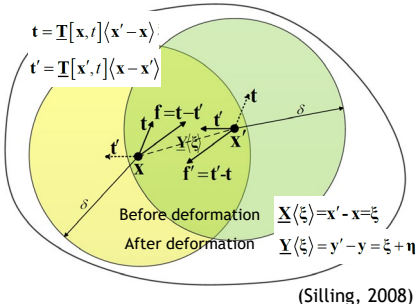
② 研究方法

③ 结果讨论

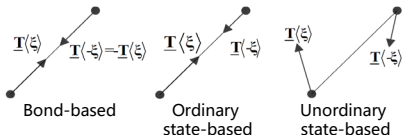
④ 应用拓展

近场动力学理论框架和微分算子

Interaction between material points in PD modeling framework



(Silling, 2008)



Peridynamic differential operators

$$\frac{\partial^{n_1+n_2+\dots+n_M} f(\mathbf{x})}{\partial x_1^{p_1} \partial x_2^{p_2} \dots \partial x_M^{p_M}} = \int_{H_{\mathbf{x}}} f(\mathbf{x} + \xi) g_N^{p_1 p_2 \dots p_N}(\xi) dV_{\xi}$$

(Madenci et al., 2016)

Governing equations
of free surface flow

→
PDDO

Non-local form
expression

Lagrangian description

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= -\rho \nabla \cdot \mathbf{v} \\ \rho \frac{\partial \mathbf{v}}{\partial t} &= -\nabla P + \mathbf{F}_V + \mathbf{F}_B + \mathbf{F}_S \end{aligned}$$

where

$$P = \frac{\rho_0 c_f^2}{\gamma} \left(\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right) + P_0$$

$$\mathbf{F}_V = \mu(\Delta \mathbf{v})$$

$$\mathbf{F}_B = 0$$

$$\mathbf{F}_S = \beta \kappa \nabla s$$

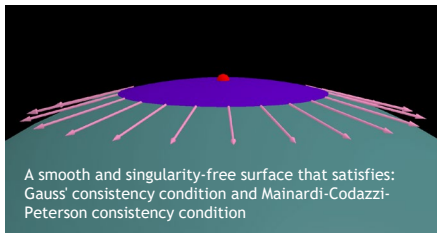
PD integral equations

$$\begin{aligned} \frac{\partial \rho(\mathbf{x})}{\partial t} &= -\rho(\mathbf{x}) \int_{H_{\mathbf{x}}} \mathbf{g}_1(\xi) \cdot (\mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})) dV_{\xi} \\ \rho \frac{\partial \mathbf{v}}{\partial t} &= - \int_{H_{\mathbf{x}}} (P(\mathbf{x}') - P(\mathbf{x})) \mathbf{g}_1(\xi) dV_{\xi} \\ &\quad + \int_{H_{\mathbf{x}}} \mu(\mathbf{x}) (\mathbf{v}(\mathbf{x}')) \\ &\quad - \mathbf{v}(\mathbf{x}) \text{tr}(\mathbf{g}_2(\xi)) dV_{\xi} \\ &\quad - \beta \int_{H_{\mathbf{x}}} \kappa (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) dV_{\xi} \end{aligned}$$

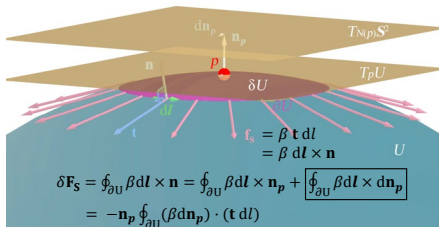
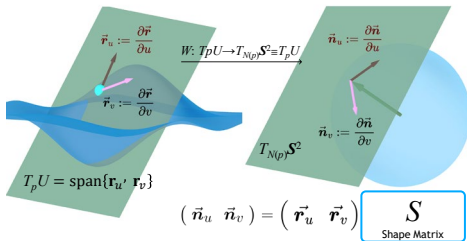
where

$$\kappa = \frac{\int_{H_{\mathbf{x}}} (\mathbf{n}(\mathbf{x}') - \mathbf{n}(\mathbf{x})) \cdot \mathbf{g}_1(\xi) dV_{\xi}}{\int_{H_{\mathbf{x}}} w_0(\|\xi\|) dV_{\xi} / \int_{H_{\mathbf{x}}} w_0(\|\xi\|) dV_{\xi}}$$

界面流体元的表面张力模型

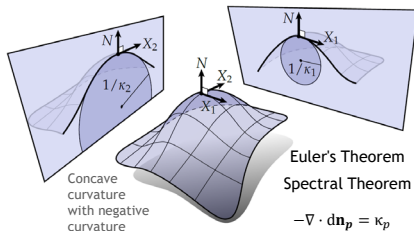


$$\vec{r}(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \\ z(u, v) \end{pmatrix} \xrightarrow{N: U \rightarrow S^2 \subset \mathbb{R}^3} \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|} \quad \boxed{\vec{n}(u, v)} = \begin{pmatrix} n_1(u, v) \\ n_2(u, v) \\ n_3(u, v) \end{pmatrix}$$



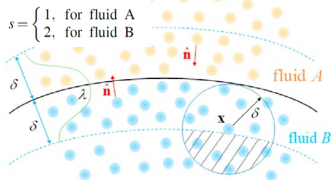
$$\text{Gauss theorem} \Rightarrow \mathbf{n}_p \oint_{\partial U} (\beta d\mathbf{n}_p) \cdot (\mathbf{t} d\mathbf{l}) = \beta (\nabla \cdot d\mathbf{n}_p) \delta U \mathbf{n}_p$$

$$\mathbf{F}_S = \lim_{\delta U \rightarrow 0} \frac{\delta F_S}{\delta U} = \beta (-\nabla \cdot d\mathbf{n}_p) \mathbf{n}_p$$



表面张力计算、控制方程、时间差分格式

The non-local form of the surface tension formula



$$s(x') - s(x) = \begin{cases} \frac{2\rho_{x,0}}{\rho_{x,0} + \rho_{x,\rho}}, & \text{if } \mathbf{x} \text{ and } \mathbf{x}' \text{ belong to different phases} \\ 0, & \text{if } \mathbf{x} \text{ and } \mathbf{x}' \text{ belong to the same phase} \end{cases}$$

Introduce the truncation function to avoid the unit normal vector field having incorrect directions

$$\mathbf{n}(\mathbf{x}) = \begin{cases} \frac{\int_{H_x} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) dV_{x'}}{\left\| \int_{H_x} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) dV_{x'} \right\|}, & \text{if } \|\nabla s\| > 1.0 \times 10^{-2} \Delta x \\ 0, & \text{else} \end{cases}$$

$$\kappa^*(\mathbf{x}) = -\nabla \cdot \mathbf{n}(\mathbf{x}) = -\int_{H_x} (\mathbf{n}(\mathbf{x}') - \mathbf{n}(\mathbf{x})) \cdot \mathbf{g}_1(\xi) dV_{x'}$$

Introduce correction factors to correct the curvature field of boundary particles let $\mathbf{N}(\mathbf{x}) = \begin{cases} 1, & \text{if } \|\nabla s\| > 1.0 \times 10^{-2} \Delta x \\ 0, & \text{else} \end{cases}$

$$\zeta^*(\mathbf{x}) = \frac{\int_{H_x} \min(\mathbf{N}(\mathbf{x}), \mathbf{N}(\mathbf{x}')) w_0(\|\xi\|) dV_{x'}}{\int_{H_x} w_0(\|\xi\|) dV_{x'}}, \kappa^{**}(\mathbf{x}) = \frac{\kappa^*(\mathbf{x})}{\zeta^*(\mathbf{x})}$$

(Gao & Oterkus, 2020a)

Continuity Equation

$$\rho_i^{n+1} = \rho_i^n - \rho_i^n \Delta t \sum_{j=1}^{N_i} [(\mathbf{v}_j^n - \mathbf{v}_i^n) \cdot \mathbf{g}_1(\xi_{ij}^n)] V_j$$

$$\rho_i^{n+1} = \rho_i^n - \frac{2\Delta x^2}{\pi \delta^2} \rho_i^n \Delta t \sum_{j=1}^{N_i} (\mathbf{v}_j^n - \mathbf{v}_i^n) \frac{[\xi_{ij1} \quad \xi_{ij2}]^T}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|}$$

Momentum equation

$$\mathbf{a}_i^{n+1} = \frac{1}{\rho_i^{n+1}} \left(\begin{aligned} & -\sum_{j=1}^{N_i} (P_j^n - P_i^n) \mathbf{g}_1(\xi_{ij}^n) V_j + \sum_{j=1}^{N_i} \mu_i (\mathbf{v}_j^n - \mathbf{v}_i^n) \text{tr}(\mathbf{g}_2(\xi_{ij}^n)) V_j \\ & -\beta \frac{(\sum_{j=1}^{N_i} \min(N_i, N_j) (\mathbf{n}_j^n - \mathbf{n}_i^n) \cdot \mathbf{g}_1(\xi_{ij}^n) V_j)}{\sum_{j=1}^{N_i} \min(N_i, N_j) w_0(\|\xi_{ij}^n\|) V_j / \sum_{j=1}^{N_i} w_0(\|\xi_{ij}^n\|) V_j} \left(\sum_{j=1}^{N_i} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi_{ij}^n) V_j \right) \end{aligned} \right) + \frac{1}{\rho_i^{n+1} \mathbf{F}^B}$$

$$\mathbf{a}_i^{n+1} = \frac{2\Delta x^2}{\pi \delta^2 \rho_i^{n+1}} \left(\begin{aligned} & -\sum_{j=1}^{N_i} (P_j^n - P_i^n) \frac{[\xi_{ij1} \quad \xi_{ij2}]^T}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|} + \frac{3\mu_i}{\delta} \sum_{j=1}^{N_i} \frac{(\mathbf{v}_j^n - \mathbf{v}_i^n)}{\sqrt{\|\xi_{ij1}^2 + \xi_{ij2}^2\|}} \\ & - \frac{(\sum_{j=1}^{N_i} \min(N_i, N_j) (\mathbf{n}_j^n - \mathbf{n}_i^n) \cdot \mathbf{g}_1(\xi_{ij}^n) V_j) \left(\sum_{j=1}^{N_i} (s(\mathbf{x}') - s(\mathbf{x})) \frac{[\xi_{ij1} \quad \xi_{ij2}]^T}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|} \right)}{\frac{\pi \delta^2}{2\beta \Delta x^2} \sum_{j=1}^{N_i} \min(N_i, N_j) w_0(\|\xi_{ij}^n\|) V_j / \sum_{j=1}^{N_i} w_0(\|\xi_{ij}^n\|) V_j} \end{aligned} \right) + \frac{1}{\rho_i^{n+1} \mathbf{F}^B}$$

Time difference scheme

Second-order Verlet-Velocity algorithm

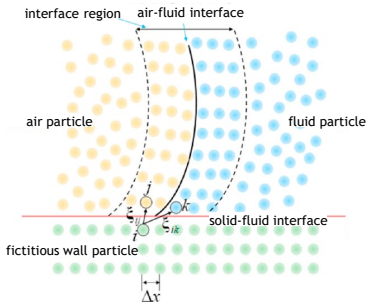
$$\begin{cases} \mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \frac{1}{2} (\mathbf{a}_i^{n+1} + \mathbf{a}_i^n) \Delta t \\ \mathbf{u}_i^{n+1} = \mathbf{u}_i^n + \mathbf{v}_i^n \Delta t + \frac{1}{2} \mathbf{a}_i^n \Delta t^2 \\ \mathbf{x}_i^{n+1} = \mathbf{x}_i^n + \mathbf{u}_i^{n+1} \end{cases}$$

Courant-Friedrichs-Levy stability condition

$$\Delta t < \min \left(\frac{h_{\text{smooth}}}{|\mathbf{v}_{\text{max}}| + c_B}, \frac{1}{8} \frac{h_{\text{smooth}}^2}{\max(\mu/\rho_0)}, \frac{1}{4} \sqrt{\frac{h_{\text{smooth}}}{g}}, \frac{1}{4} \sqrt{\frac{\min(\rho_0) h^3}{2\pi\beta}} \right)$$

For two-dimensional problems, the relative value form of the near-field dynamic differential operator is adopted.

边界的处理



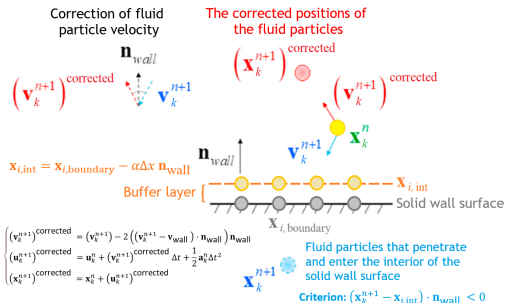
Considering the no-slip boundary, the fluid velocity is zero at the wall surface.

$$\mathbf{v}_i^n = 2\mathbf{v}_{\text{wall}} - \sum_{k=1}^{N_f} w_0(\|\boldsymbol{\xi}_{ij}^n\|) \mathbf{v}_k^n$$

$$P_i = \frac{\sum_{k=1}^{N_f} (P_k + (\mathbf{F}_k^B - \rho_k \mathbf{a}_{\text{wall}}) \cdot \boldsymbol{\xi}_{ik}^n) w_0(\|\boldsymbol{\xi}_{ij}^n\|)}{\sum_{k=1}^{N_f} w_0(\|\boldsymbol{\xi}_{ij}^n\|)}$$

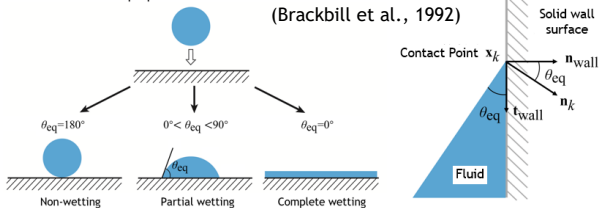
$$\rho_i = \rho_{0,i} \left(\frac{P_i - P_{\text{ref},i}}{P_{\text{ref},i}} + 1 \right)^{1/\gamma_i}$$

(Gao & Oterkus, 2020a)

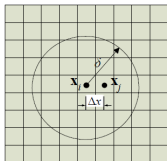


Consider the surface properties of the solid

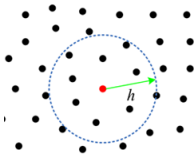
(Brackbill et al., 1992)



空间离散（近场域搜索）和自由液面识别

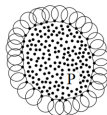


Two-dimensional structure with central orthogonal uniform discretization.

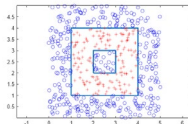


By calling the library function range search to conduct a KD-tree neighborhood search for the specified particle.

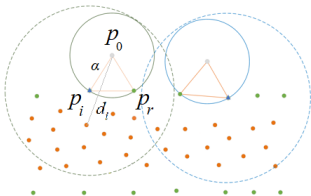
The free surface is identified by calling library functions such as:



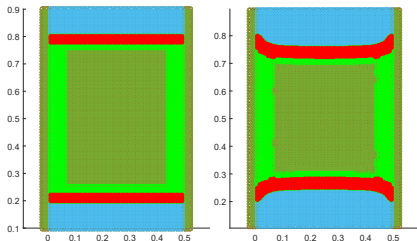
alphaShape



inpolygon

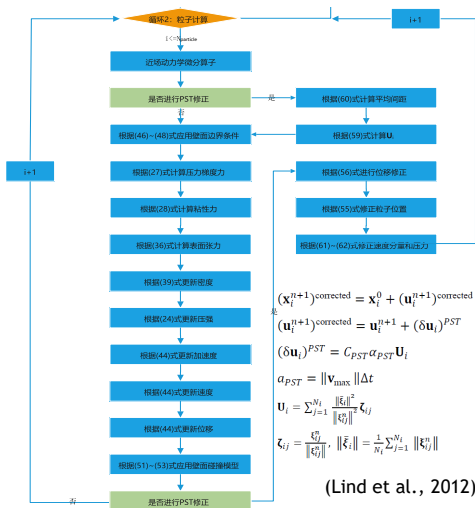


对所有流体粒子和固体粒子构成的结构使用alpha shapes算法，可以找到该结构的边界点。再使用inpolygon函数剔除背景中多余空气粒子！



防聚集的粒子迁移技术

- PST (Particle Shifting Technology) 技术通过引入位移修正来调整粒子位置，使粒子在移动后保持近似均匀分布



$$\begin{aligned}
 (\mathbf{v}_{i1}^{n+1})^{\text{corrected}} &= \left[\mathbf{v}_{i1}^{n+1} + \frac{2\Delta x^2}{\pi\delta^2} \sum_{j=1}^{N_i} (\mathbf{v}_{j1}^n - \mathbf{v}_{i1}^n) \frac{[\xi_{ij1} \quad \xi_{ij2}]^T}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|} \cdot (\delta \mathbf{u}_i)^{PST} \right. \\
 &+ \frac{3\Delta x^2}{2\pi\delta^3} \sum_{j=1}^{N_i} ((\delta \mathbf{u}_i)^{PST})^T \cdot (\mathbf{v}_{j1}^n - \mathbf{v}_{i1}^n) \frac{\begin{pmatrix} 3\xi_{i1}^2 - \xi_{i2}^2 & 4\xi_{i1}\xi_{i2} \\ 4\xi_{i1}\xi_{i2} & -\xi_{i1}^2 + 3\xi_{i2}^2 \end{pmatrix}}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|^{3/2}} \cdot (\delta \mathbf{u}_i)^{PST}, \\
 \mathbf{v}_{i2}^{n+1} &+ \frac{2\Delta x^2}{\pi\delta^2} \sum_{j=1}^{N_i} (\mathbf{v}_{j2}^n - \mathbf{v}_{i2}^n) \frac{[\xi_{ij1} \quad \xi_{ij2}]^T}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|} \cdot (\delta \mathbf{u}_i)^{PST} \\
 &\left. + \frac{3\Delta x^2}{2\pi\delta^3} \sum_{j=1}^{N_i} ((\delta \mathbf{u}_i)^{PST})^T \cdot (\mathbf{v}_{j2}^n - \mathbf{v}_{i2}^n) \frac{\begin{pmatrix} 3\xi_{i1}^2 - \xi_{i2}^2 & 4\xi_{i1}\xi_{i2} \\ 4\xi_{i1}\xi_{i2} & -\xi_{i1}^2 + 3\xi_{i2}^2 \end{pmatrix}}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|^{3/2}} \cdot (\delta \mathbf{u}_i)^{PST} \right]^T
 \end{aligned}$$

$$\begin{aligned}
 (P_i^{n+1})^{\text{corrected}} &= P_i^{n+1} + \frac{2\Delta x^2}{\pi\delta^2} \sum_{j=1}^{N_i} (P_j^n - P_i^n) \frac{[\xi_{ij1} \quad \xi_{ij2}]^T}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|} \cdot (\delta \mathbf{u}_i)^{PST} \\
 &+ \frac{3\Delta x^2}{2\pi\delta^3} \sum_{j=1}^{N_i} ((\delta \mathbf{u}_i)^{PST})^T \cdot (P_j^n - P_i^n) \frac{\begin{pmatrix} 3\xi_{i1}^2 - \xi_{i2}^2 & 4\xi_{i1}\xi_{i2} \\ 4\xi_{i1}\xi_{i2} & -\xi_{i1}^2 + 3\xi_{i2}^2 \end{pmatrix}}{\|\xi_{ij1}^2 + \xi_{ij2}^2\|^{3/2}} \cdot (\delta \mathbf{u}_i)^{PST}
 \end{aligned}$$

速度和压力也可以根据位移的修正量进行修正
(二维完整对称近场域下相对值形式)

(Gao & Oterkus, 2020a)

① 研究背景

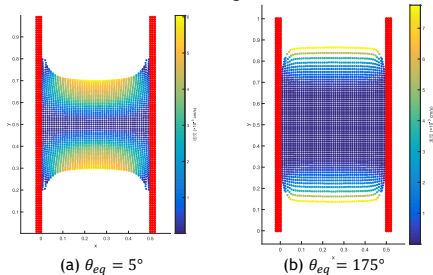
② 研究方法

③ 结果讨论

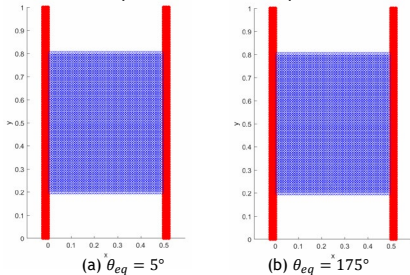
④ 应用拓展

计算结果

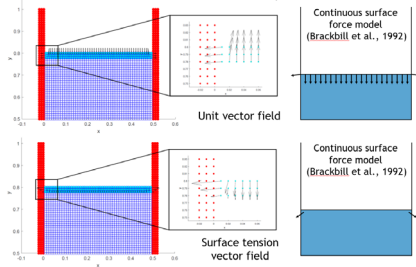
Final configuration



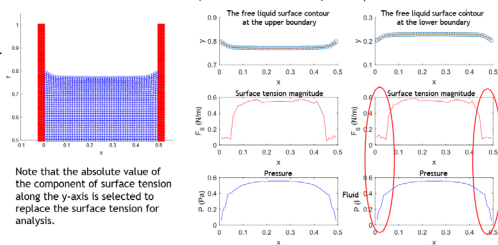
Liquid level evolution process



Initial configuration ($\theta_{eq} = 5^\circ$)

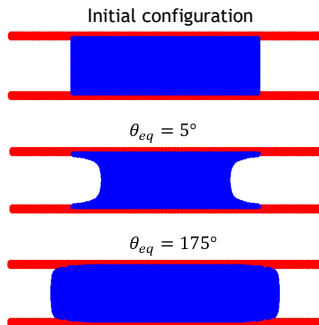


Particle distribution at 7500 time steps for the concave liquid surface example ($\theta_{eq} = 5^\circ$)

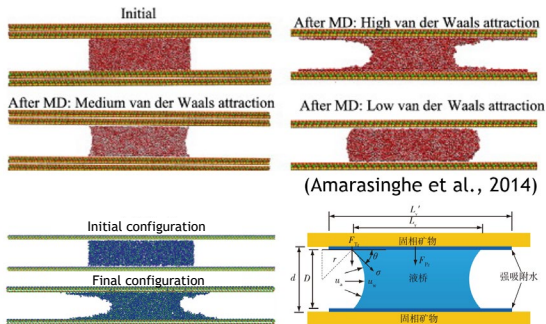


算例比较 (PD vs. MD)

The stronger the adsorption of the solid boundary on the pore water, the more "concave" the curved liquid surface will be; conversely, it will be more "convex".



(a) PD results



(Luo et al., 2022)

(b) MD results

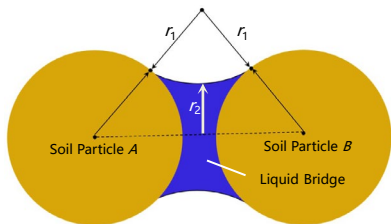
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③ 结果讨论

④ 应用拓展

非饱和黏土弯液面演化的数值模拟

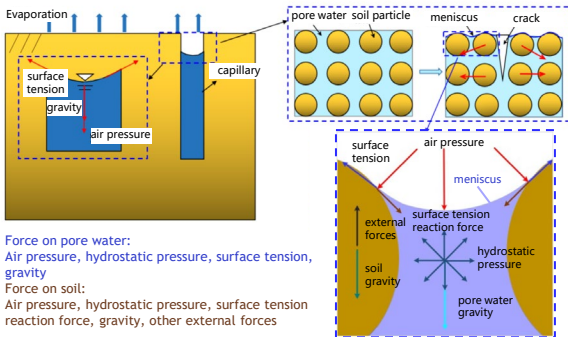


The matrix suction of unsaturated soil is defined by Young-Laplace Equation, related to the morphology curved liquid surface

$$s = \gamma_s \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

找到黏土干缩开裂与微观液桥结构的联系是关键！

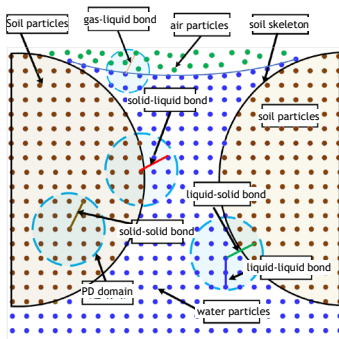
Micro-scale meniscus evolution controls capillary suction
→ governs macro-scale shrinkage and cracking!



“热-水-力”耦合作用下弯液面的演化探究

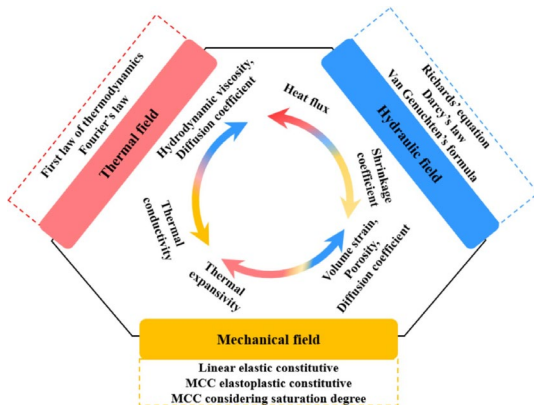
Simplified single liquid bridge model

Capillary action ✓
Evaporation effect ✗



The microscopic structure of clay

Investigation on the Evolution of Curved Liquid Surface under the Coupling Effect of Heat, Water and Force is our future work!

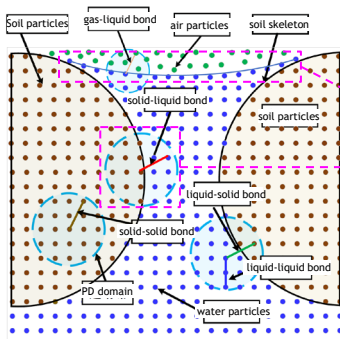


“热-水-力”耦合作用下弯液面的演化探究

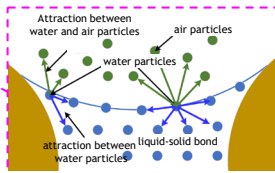
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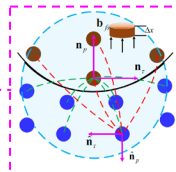


One-way gas-liquid interaction

Boundary conditions:

Gas $\rightarrow$ Liquid: Surface tension

Solid $\rightleftharpoons$ Liquid: Decomposes into two forces acting in the pressure direction and the shear direction



Fluid-solid interaction

“热-水-力”耦合作用下的数学物理建模

The basic equations of the micro-scale "solid-liquid-gas" three-phase medium of clay under the "thermal-hydrodynamic" coupling effect:

mass conservation (continuity) equation

$$\frac{D\rho_l(T)}{Dt} + \rho_l(T)\nabla \cdot \mathbf{U}_l = \rho_l(T)\dot{m} \left(\frac{1}{\rho_g(T)} - \frac{1}{\rho_l(T)} \right) \quad \frac{D\rho_s}{Dt} + \rho_s\nabla \cdot \mathbf{U}_l = 0$$

momentum conservation equation

$$\frac{D(\rho_l(T)\mathbf{U}_l)}{Dt} = -\nabla p(T) + \mathbf{F}_l^v + \mathbf{F}_l^B + \mathbf{F}_l^{sl} \quad m_s \frac{\partial \mathbf{U}_s}{\partial t} = \mathbf{F}_s^{pv} + \mathbf{F}_s^B + \mathbf{F}_s^{ls} + \mathbf{F}_s^S$$

water vapor diffusion equation

$$\frac{\partial Y}{\partial t} + \nabla \cdot (Y\mathbf{U}_l) = \nabla \cdot (D_v(T)\nabla Y)$$

heat energy conservation equation

$$\frac{\partial(\rho_l c_{lp} T_l)}{\partial t} + \nabla \cdot (\rho_l c_{lp} \mathbf{U}_l T_l) = \nabla \cdot (k_{lT} \nabla T_l) - \dot{m} h_{ev} + \frac{1}{2} \mu(T_l) \mathbf{S} : \mathbf{S}$$

heat transfer equation

$$\frac{\partial(\rho_s c_{sp} T_s)}{\partial t} = \nabla \cdot (\overset{\text{conduction}}{k_{sT}} \nabla T_s) \quad k_{sT} \frac{\partial T_s}{\partial n_{sl}} = \overset{\text{convection}}{\alpha_{sl}} (T_s - T_l)$$

谢谢！

Thanks for your time and attention!